IWS-RL: understanding the project (edition 5, evening of 2 Nov 2026)

Repo: https://github.com/adithyx-0/iws-rl | Plan for the next 30 hours: `docs/PLAN_v1.md` | Full history: `docs/PROJECT_LOG.md`

This document explains, for anyone on the team, what we are doing, why, where every idea came from, what we found, and what is still open. It is updated at each real milestone (see the update log at the end).

1. The project in one paragraph

In many cities (e.g. Indian cities) piped water arrives only a few hours per day. A utility opens valves zone by zone. Households keep roof tanks to survive the gaps, and the utility cannot see how full those tanks are. We built a simulator (EPANET hydraulics via the WNTR package, network Net3, 4 zones) and trained reinforcement-learning agents to decide every hour which zones get water, aiming to serve as much demand as possible while protecting the worst-off zone (equity). We compare the agents with simple rules, an optimised fixed daily schedule, and (soon) a clairvoyant oracle.

2. How we got here (chronology)

1. Early October: the first project was SafeCharge-RL (EV charging). It is finished to a large extent and pushed on GitHub, but the faculty said EV charging is too common.
2. 30 Oct: we shelved it and searched for a new topic. An earlier Claude had written candidate files (`01-07`, about a *different* project: exam-aware study scheduling). I eliminated all 12 candidates against the other teams' topics (Excel sheet from the course).
3. 30-31 Oct: you brought `iws-rl.zip` (intermittent water supply), built by another Claude (Opus). We trained PPO, DQN and ablations and ran a full evaluation (v1 results).
4. 1 Nov: we found the v1 environment was too easy (see section 5), built the harder v2 environment, put everything on GitHub, and an external review (`docs/reference/improvements_for_ref.md`) sharpened the plan.
5. 1-2 Nov: three-laptop sprint (`docs/PLAN_v1.md`) to finish and present.

3. Where each idea came from

Idea / component	Source	Status
Topic: equity-aware RL for intermittent water supply (IWS)	Other Claude (Opus) produced `iws-rl.zip`; we chose it after eliminating 12 earlier candidates	Adopted; faculty approval stated as obtained by the user
EPANET hydraulics, network Net3	WNTR package (bundled example network), https://github.com/USEPA/WNTR	Tool
PPO, DQN, A2C implementations	Stable-Baselines3, https://github.com/DLR-RM/stable-baselines3	Tool
Household roof tanks, daily budget, equity reward	Original `iws-rl.zip` design; values (tank 0.3-0.8 days, alpha=4) are **assumptions**	Needs cited ranges (Adithyan)
Privileged greedy (reads hidden tank levels)	Added by us as an upper-ish reference	Not a true oracle (scores below PPO on v2); the true clairvoyant oracle is `oracle.py`
Optimised fixed daily schedule (CE search)	Added by us as the stand-in for Bayesian-optimisation schedules in the literature (Hendrickson & Sela)	Retuned (40 iterations, population 60, 3 seeds); it is CE search, not Bayesian optimisation
Diagnosis: scarce supply is degenerate	Our own analysis (`scarce_diag.py`, `scarce_trace.py`)	Done
Hydraulics never bind in v1; demand_scale=8 gives v2	Our own probes (`hyd_probe.py`, `scale_study.py`); the 4-8x peak-flow argument is our reasoning, not a citation	Done
Supply ceiling (mean supply can't exceed ~0.773 in v1)	External review `docs/reference/improvements_for_ref.md`; consistent with our scarce result 0.4754 vs ceiling 0.473	Consistent; not independently re-derived by us

Idea / component	Source	Status
Adaptive policy: domain randomisation + recurrent (LSTM) policy	Same review; resembles rapid-motor-adaptation ideas in robotics (from memory, **verify before citing**)	Done (C1): domain randomisation helps under shift; LSTM did not
Equity as a reward weight lambda; Pareto front over lambda	Same review; Siddique et al., ICML 2020 is related fair-RL work, but our method is a reward penalty, not constrained RL and not their welfare optimisation	Done (lambda sweep)
Clairvoyant oracle for regret	Our design: env is deterministic given a seed, so optimise the full action sequence per scenario	Done (Gate 1)

4. Related work found so far

Honesty note: items were first found as titles or snippets. A later audit (`docs/reference/iws-rl-literature-audit.md`, 2 Nov, 23 queries, run by another AI session; check cited sources yourself before presenting) read in full: Hendrickson & Sela journal and conference versions, Kurian et al. 2024, Hajgato et al. 2020 and Gym4ReaL; abstract only: Kurian 2023, Siddique 2020, Holgado-Sanchez 2024, Wanjiru 2016; the rest snippet only. Part A of the verification plan (`docs/results-adithya-paper.md`, 2 Nov evening) has since read in full: Holgado-Sanchez 2024, Wanjiru 2016, Kurian 2023, Solgi 2015, Ayyash 2024, Ansalone 2025. Still abstract-only: Gullotta 2021 (valves), CORE, Ramani, Ameyaw. That verification was itself AI-assisted, so a human must still find the sentence for every fact shown on a slide. Part B (Adithyan) is still open.

Closest to our problem (non-RL, equity in intermittent supply)

- Hendrickson & Sela, 2024: Bayesian optimisation of flow-control valve schedules for local supply and global equity, volume-driven demand with private tanks, 6 zones. https://doi.org/10.1016/j.scs.2024.105615 ; conference version https://doi.org/10.3390/engproc2024069075 (Eng. Proc. 69(1):75, with a third author; 69/1/184 is Kurian et al. 2024 and 69/1/7 is unrelated: verified by the audit, full text read)
- 2023 Water Supply, MINLP minimising supply inequity under hydraulic constraints: https://doaj.org/article/37953cf59e474f7687beb590577f3458
- Kurian et al. 2023, Water Supply, equitable supply in intermittently operated rural schemes: https://iwaponline.com/ws/article/23/11/4520/98117/Equitable-supply-in-intermittently-operated-rural
- UCSB intermittent-supply scheduling thesis: https://escholarship.org/uc/item/3xb4j737
- Bengaluru thesis on inequity under intermittent supply: https://etd.iisc.ac.in/handle/2005/5925
- Delhi/Bengaluru practice and equity discussion: https://www.indiawaterportal.org/health-and-sanitation/urban-sanitation/intermittent-water-distribution-networks-tale-two-cities

RL for water networks (no equity, not intermittent supply)

- Hu et al. 2020, deep RL valve scheduling for contamination isolation: https://aimspress.com/aimspress-data/mbe/2020/1/PDF/mbe-17-01-006.pdf
- Multi-agent RL (MADDPG) for pump and valve scheduling: https://bohrium.dp.tech/paper/arxiv/2408.08442
- PPO for pump scheduling on a real network with 18 demand zones: https://arxiv.org/pdf/2310.09412
- DRL-EPANET (NeurIPS workshop 2022): https://mlanthology.org/neuripsw/2022/belfadil2022neuripsw-drlepanet
- RL for household storage and tariffs in smart water networks (IFIP Networking 2025): https://dl.ifip.org/db/conf/networking/networking2025/1571132384.pdf
- Deep RL for pumps in water distribution systems: https://arxiv.org/pdf/2010.06460
- Safe RL control for water networks (Aalborg): https://vbn.aau.dk/en/publications/safe-reinforcement-learning-control-for-water-distribution-networ-2

Methods we may use (from the external review; not yet verified by us)

- Fair RL with generalised Gini welfare (Siddique et al., ICML 2020): https://arxiv.org/pdf/2008.07773

- Expected-welfare training does not guarantee fairness (two 2026 arXiv papers): https://arxiv.org/pdf/2606.18111 and https://arxiv.org/pdf/2608.00175
- Past-discounting for Markovian fairness: https://arxiv.org/abs/2504.01154v2
- Asymmetric actor-critic / teacher-student with privileged information (Cai et al., NeurIPS 2024): https://arxiv.org/abs/2412.00985
- CVaR in RL (Tamar et al.): https://arxiv.org/pdf/1406.3339v3.pdf ; Predictive CVaR policy gradient: https://rlj.cs.umass.edu/2024/papers/RLJ_RLC_2024_81.pdf
- Safe RL tooling: Safety-Gymnasium https://arxiv.org/pdf/2310.12567 ; OmniSafe https://github.com/PKU-Alignment/omnisafe
- Intermittent supply background (Dehra Dun): https://sswm.info/node/1445

Is it new? (audit result, 2 Nov)

Answer: **unclear (confidence low-medium)**. No paper was found combining RL, intermittent-supply zone valve control, an equity term and household storage, but absence in 23 web queries is not proof, and the search could not use Scholar, IEEE, Scopus or ScienceDirect directly. Closest prior work: Hendrickson & Sela 2024 (open-loop Bayesian-optimised valve schedules for equity with tanks in WNTR; no RL; they test fixed schedules against +-50% tank-size changes but only versus no control), Kurian et al. 2023 (MINLP schedule for equity), Holgado-Sanchez et al. 2024 (constrained RL, equity, drought; abstract only), Wanjiru et al. 2016 (open-loop vs closed-loop MPC for household tanks; already states the qualitative regime result, so the idea is not new, only our RL evidence on this network). RL for water without equity or intermittent supply: Hu et al. 2023, Hajgato et al. 2020 (demand randomisation), Gym4ReaL 2025 (EPANET, binary pump actions). "Cai et al. 2024" and an "npj Clean Water" equity paper could not be found; do not cite them.

Allowed wording: "To our knowledge, based on a targeted search, this is an exploratory study applying RL to equity-aware valve scheduling in intermittent supply; adaptive control helped under the shifts we tested, not in-distribution." Not allowed: "first/novel" without the qualifier, "RL beats optimised schedules" without "under the shifts we tested", "constrained/Lagrangian RL", "robust" in general, "statistically significant" without stating seeds and CIs, results transfer to real networks. Note also that our household tanks sit outside the EPANET hydraulics (a simplification relative to Hendrickson & Sela).

Verification results, Part A (2 Nov evening; summary of `docs/results-adithya-paper.md`*)*

- **Holgado-Sanchez et al. 2024 is constrained RL for equity in drought water allocation** (tabular Double Q-learning, Gini-based equity reward, RCPO-style Lagrangian with threshold 0.7). Its setting is a tanker truck serving 4 villages: no pipe network, valves, zones, household tanks or intermittent supply; no fixed-schedule baseline; no demand-shift tests. So never write "no RL for equity in water". Safe wording: prior RL for equitable allocation under drought uses a tanker-truck abstraction, not a hydraulic network with zonal valves and household tanks.
- **Hendrickson & Sela 2024 (journal)**: open-loop Bayesian-optimised valve schedules, no tank sensing assumed; they argue sensor-based central control may be impractical in intermittent supply. We must justify our closed-loop observations (tank-level information). Their tanks sit inside the EPANET model (ours outside); their baseline tank is 1 day of demand (ours 0.3-0.8); their equity metric is a mean-absolute-deviation index (ours is minimum-zone supply). Their schedule stays better than no control at +-50% tank size.
- **Wanjiru et al. 2016**: one household rooftop tank, energy-cost objective, not intermittent supply. Open-loop copes with random noise (even 50%) but fails under a +70% demand spike; closed-loop MPC fixes it at about 9.5% higher cost. Supports our "fixed schedule is fine until the pattern changes" story only as an analogy.
- **Kurian et al. 2023**: MINLP, reformulated as an integer program; village-level demand nodes, no household tank dynamics, daily open loop, no RL.
- **Ansalone et al. 2025** (corrected): "A Reinforcement Learning Approach to Demand Balancing and Tariff Optimization in Blockchain-Based Smart Water Networks", four authors, IFIP Networking 2025; DQN on tariffs and consumption patterns, continuous supply, no valves or equity.

- **Assumption sources**: tank size 0.3-0.8 days and surge alpha=4 have no direct source (disclose as assumptions). Pressure 7-30 m and reference 15/20 m are citable (Ayyash 2024, Solgi 2015, Kurian 2024). Supply 50-83% of demand has support (Solgi, Gullotta, Ayyash); the 90% end is an assumption.
- **Possibly missing from the audit (to check)**: Abdelazeem & Meyer 2024 (JWRPM 150(1)); Randeniya et al. 2022 (Water Supply 22(5)); Solgi et al. 2020; Nyahora et al. 2020; Ilaya-Ayza et al. 2017.

Final M1 results (2 Nov morning): tuned fixed schedule, ablations, fairness Pareto

Source files: `results_v2/summary.csv`, `results_v2/paired_vs_ce_tuned.csv`, figures in `results_v2/figs/` (`pareto.png`, `regime_map.png`). Script: `analyze_final.py`. CIs are paired bootstraps over test scenarios only; training-seed variance is excluded. PPO, PPO+DR, DQN and no-equity have 5 seeds each, tuned schedule 3 (CE 40 iterations, population 60), lambda sweep 2-3 per point (after the 2 Nov seed top-up).

Condition	PPO	PPO+DR	DQN	No-equity	Tuned schedule
default (return)	-9.67	-9.92	-10.02	-10.14	-9.15
alpha=2 (demand surge, shifted)	-9.67	-9.15	-8.80	-9.66	-10.81
small tanks (shifted)	-17.77	-17.57	-16.48	-18.55	-17.62
alpha=8	-11.18	-10.90	-11.79	-11.62	-10.98

- **The tuned fixed schedule beats plain PPO in the normal setting** (paired, significant), and in most conditions on both return and worst-zone supply. Our earlier "PPO beats the fixed schedule" was an artefact of an under-tuned baseline.
- **RL significantly beats the tuned schedule where conditions shift**: alpha=2 (all RL methods, return and worst-zone), small tanks (DQN on return; PPO+DR ties), scarce (return, DQN/PPO/PPO+DR). A fixed schedule cannot react to these.
- **Fairness weight lambda** (PPO, default): worst-zone 0.573 (lambda 0) to 0.635 (lambda 4) for a mean-supply cost of 0.010; node-level Gini does not improve. Removing the equity term lowers worst-zone supply (0.564 vs 0.601).
- **Honest limits:** tuned schedule overfits its training scenarios somewhat (train -7.4, test -9.15); single network; assumed parameters; the lambda points have 2 seeds; the novelty claim is unverified (literature audit pending).
- **Story:** RL is not better everywhere. It matters when the system differs from what a schedule was tuned for, and the equity weight gives a controllable fairness trade-off.

6. Open problems and the plan

- Fairness Pareto curve (C2) not finished: lambda sweep running on the MacBook, the Vivobook has the main version.
- No method contribution beyond environment + PPO -> C1 adaptive policy (domain randomisation + LSTM), C2 fairness Pareto front, C3 regime map (see `docs/PLAN_v1.md`).
- Fixed-schedule baseline is under-tuned and wrongly named (cross-entropy search, not Bayesian optimisation).
- No learning curves; single network; tank sizes and surge factor are assumptions; everything is simulation.
- Related work not yet read in full; faculty approval of the topic not yet confirmed.

7. Where things are in the repo

- `README.md` start here; `docs/PROJECT_LOG.md` chronological findings; `docs/PLAN_v1.md` the three-laptop plan; `docs/GIT_GUIDE.md` git for beginners; `docs/reference/improvements_for_ref.md` the external review.
- Code: `iws_env.py` (environment), `baselines.py`, `train_variant.py`, `eval_all.py`, `bo_schedule.py`, `v2cfg.py` (set `IWS_V2=1` for the harder environment), `run_jobs.py` (cross-platform job queue).
- Results: `results/` (v1), `results_v2/` (v2), models in `models/` and `models_v2/`.

8. Update log

Date	What changed
1 Nov 2026	Edition 1: project history, sources, v1 and v2 results, plan v1
night of 1 Nov	Edition 2: oracle and Gate 1, domain-randomisation and LSTM results

- 2 Nov morning (edition 3): added tuned-schedule baseline, no-equity/DQN runs, lambda Pareto, paired CIs; main conclusion changed (see section 5, final M1 results).
- 2 Nov afternoon (edition 4): seed top-up (5 seeds for PPO+DR, DQN, no-equity), literature audit folded in (citation fixes, novelty wording), paper verification plan `docs/VERIFY_PAPERS.md`.
- 2 Nov evening (edition 5): Part A paper verification folded in (Holgado-Sanchez is constrained RL for equity in a tanker-truck setting; H&S assume no tank sensing; Ansalone citation corrected; assumption sources listed). Part B still open.